

HCI HAPPY ENGINE

CROSS-ANALYSIS SPEC V1

PURPOSE

The HAPPY Engine must not rely on direct questionnaire scores alone.

The strongest insights emerge when subjective athlete perception is crossed with objective HCI performance data.

LEVEL V1 — DIRECT READING

Uses only athlete responses.

Examples:

CONFIDENCE

NERVE

HEED

ALIGNMENT

PLAN

POWERING

YEARN

Limitation:

Good for screening.

Weak for diagnosis.

LEVEL V2 — INTERNAL HAPPY CROSSING

Cross HAPPY and COMANDO variables.

Examples:

CONFIDENCE LOW + POWERING HIGH

= Athlete is working hard but lacks belief.

NERVE LOW + HEED LOW

= Athlete struggles with pressure and is unaware of arousal regulation.

PLAN LOW + ALIGNMENT LOW

= Coach-athlete direction problem.

YEARN HIGH + ORGANIZATION LOW

= High ambition with poor execution structure.

Outcome:

Explains WHY a score is low.

LEVEL V3 — HAPPY + HCI PERFORMANCE

Cross subjective data with objective HCI engines.

HAPPY

+ TARGETS

+ STRUCTURE

+ RHYTHM

+ RESULTS

Main Crossings:

CONFIDENCE × RESULT

Confidence supported by evidence?

CONFIDENCE × CONSISTENCY

Can confidence survive repetition?

NERVE × PRESSURE

Does perceived pressure match performance degradation?

HEED × RHYTHM

Does arousal affect rhythm?

HEED × EMOTIONAL

Does the athlete perceive emotional instability?

PLAN × PROCESS

Is the coach's plan becoming execution?

ALIGNMENT × TRANSFER

Does the athlete understand and transfer training?

POWERING × PHYSICAL

Does perceived energy match physical performance?

YEARN × TRAINING FREQUENCY

Does ambition match commitment?

ORGANIZATION × SESSION VARIABILITY

Does poor organization create inconsistent performance?

ANALYSIS × DEEPENING

Is the athlete learning or simply repeating?

OPPORTUNITY × TRAINING UTILIZATION

Is the athlete using the opportunities created by the coach?

PERCEPTION GAP MODEL

The most valuable insight is the gap between:

ATHLETE PERCEPTION

versus

OBJECTIVE EVIDENCE

Example:

CONFIDENCE = 5

CONSISTENCY = 2.4

PRESSURE = 2.1

Interpretation:

The athlete reports high confidence but objective evidence does not support it.

THREE-LAYER DASHBOARD

Layer 1

WHAT THE ATHLETE FEELS

Layer 2

WHAT THE DATA SHOWS

Layer 3

WHERE PERCEPTION AND REALITY DIVERGE

COACH INTELLIGENCE OUTPUT

The dashboard should answer:

What does the athlete feel?

What is actually happening?

What is causing the gap?

What intervention is most likely to reduce the gap?

VISION

The HAPPY Engine is not a questionnaire.

It is a perception-versus-performance intelligence system.